Spatial Population Dynamics: Fertility, Migration, and Policies*

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Abstract

We develop a parsimonious spatial model to quantify how regional policies shape population dynamics and the distribution of income and welfare. In the model, fertility and migration are jointly determined through labor and housing markets over space. We estimate the model using data on income, housing, migration, and fertility across 741 U.S. commuting zones (CZs), identifying a low elasticity of substitution between fertility and housing from the negative relationship between fertility rates and housing rents. Counterfactual simulations show that equalizing housing supply regulations to the median across CZs raises fertility in large CZs, increases aggregate population growth, and compresses the dispersion of real wages, while widening the dispersion of real housing rents.

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1 Introduction

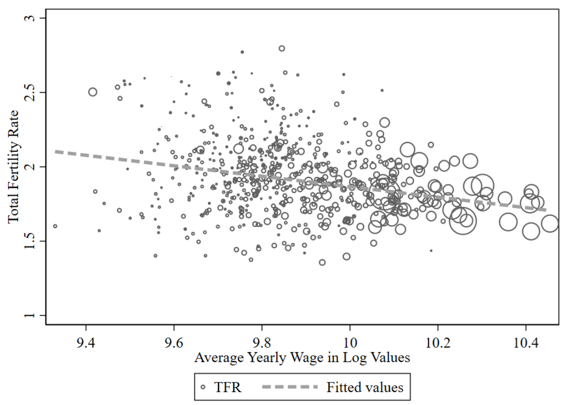
Fertility has fallen sharply across advanced economies, reshaping labor supply, the age structure of populations, and the fiscal sustainability of social insurance systems. In the United States, the total fertility rate declined from 2.1 children per woman in 1990 to 1.6 in 2023, well below the replacement level. A large body of work emphasizes changes in gender roles, social norms, labor market frictions, and the difficulty of combining careers and family life as drivers of this decline (Doepke, Hannusch, Kindermann, and Tertilt, 2023; Goldin, 2021). These explanations are important, yet they are often silent about a striking empirical regularity: fertility varies enormously across space within a country, and the lowest fertility is frequently concentrated in the most productive metropolitan areas.

This paper argues that fertility should be analyzed as the equilibrium outcome of a spatial household problem. Households choose where to live, how much housing to consume, and how many children to have. These choices interact in general equilibrium. Spatial sorting affects housing demand and local prices. It also changes the cost of raising children through both time costs—captured by wages—and space costs—captured by housing rents. In this environment, fertility is not merely a preference parameter or a reduced-form outcome. Rather, it is an endogenous choice that responds to local economic conditions, and these demographic responses feed back into the spatial distribution of population and economic activity.

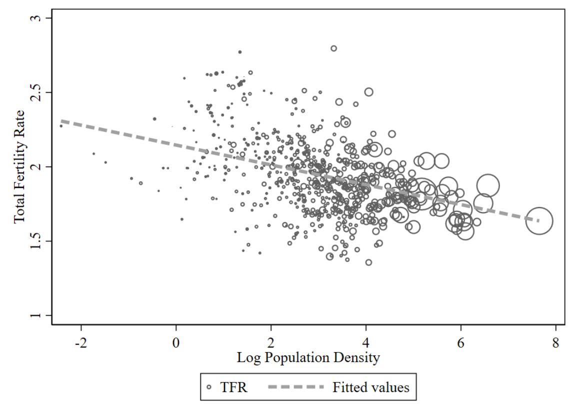
The motivation for our analysis comes from a striking set of empirical patterns. Figure 1 documents the cross-sectional relationship between fertility rates and local economic conditions across 741 U.S. commuting zones. Panel (a) shows that fertility is strongly negatively correlated with wages: a one log-point increase in average wages is associated with a 0.4 reduction in the total fertility rate. Panel (b) reveals a similar negative relationship with population density. Panels (c) and (d) show that fertility is negatively correlated with both housing prices and rents, with the rent elasticity of fertility equal to approximately -0.23. These patterns suggest that the opportunity cost of children plays a central role in determining fertility across space.

These correlations present a puzzle from the perspective of standard spatial equilibrium models. In those models, high-wage, high-rent locations offer utility levels comparable to other places, with the amenity value of living in productive cities offset by higher costs of living. But if fertility is part of households' choice set, the calculus changes: locations that offer high wages and expensive

Figure 1: Fertility Rates and Local Economic Conditions Across U.S. Commuting Zones

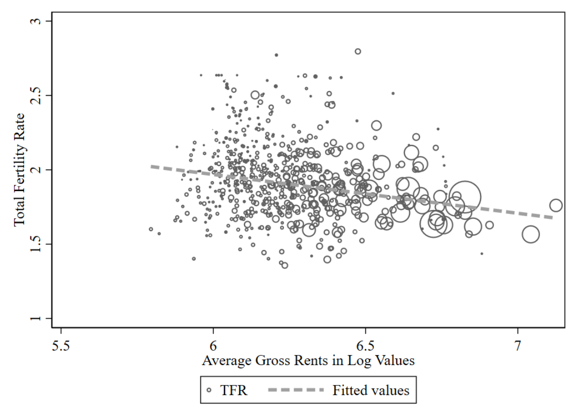


(a) Wage



(b) Population Density

(c) Housing Price



(d) Housing Rent

Notes: Data are from the 2000 U.S. Census via IPUMS USA (Ruggles, Flood, Sobek, Backman, Cooper, Drew, Richards, Rodgers, Schroeder, and Williams, 2025). Following Autor and Dorn (2013), we use IPUMS geographic identifiers to construct 1990-consistent commuting zones. We compute the total fertility rate (TFR) for women aged 15–45 using the age of youngest child in household; average annual wages are calculated from annual wage income; following Moretti (2013), we estimate housing prices and rents at the commuting zone level. The fitted line coefficients (standard errors) in panels (a)–(d) are -0.403 (0.054), -0.091 (0.006), -0.143 (0.031), and -0.227 (0.047), respectively. Circle size represents population.

housing may reduce family formation, generating demographic consequences that extend beyond the current generation. The negative correlation between productivity and fertility implies that the spatial reallocation of population toward productive cities—emphasized as a source of aggregate efficiency gains in the literature—may come at the cost of lower aggregate fertility. Understanding this tradeoff requires a framework that treats fertility as endogenous and interacts with the other margins of adjustment in the spatial economy.

We make three contributions. First, we develop a quantitative spatial equilibrium model that integrates endogenous fertility, location choice, and housing markets in a unified framework with market clearing. Unlike models that treat fertility or location in isolation, our framework captures how sorting, housing prices, and demographic dynamics interact in general equilibrium. Second, we derive analytical expressions characterizing how local economic conditions affect fertility and how these effects propagate across regions through trade, migration, and price adjustments. These results identify sufficient statistics for policy evaluation and illuminate the mechanisms underlying the spatial fertility gradient. Third, we calibrate the model to 741 U.S. commuting zones and use it to evaluate place-based fertility policies, showing how interventions targeting housing supply affect both aggregate efficiency and aggregate fertility.

Our model builds on the quantitative spatial economics framework of Redding and Rossi-Hansberg (2017), extending it to incorporate overlapping generations with endogenous fertility following Barro and Becker (1989). The economy consists of multiple regions, each producing a differentiated variety of goods. Workers are mobile across regions and face idiosyncratic location preferences, generating the smooth migration responses characteristic of discrete choice models (McFadden, 1973). Housing is supplied with region-specific elasticities, capturing the heterogeneous regulatory and geographic constraints documented by Saiz (2010). The key innovation is that households choose not only where to live and how much to consume, but also how many children to have. Children require parental time, generating an opportunity cost that rises with wages, and space, generating a cost that rises with housing rents. Crucially, we allow housing and children to be complements in preferences, so that high housing costs reduce fertility both directly and by raising the effective price of the housing-children composite.

The model yields several key insights about the mechanisms linking local economic conditions to fertility and aggregate outcomes. First, we derive closed-form expressions for the local price elasticities of fertility. When housing and children are complements—as our estimates suggest—the rent elasticity of fertility is unambiguously negative. Higher wages have an ambiguous effect, combining a negative substitution effect (children become relatively expensive) with a positive income effect. Under our calibrated parameters, the substitution effect dominates, rationalizing the negative fertility-wage correlation in the data. Second, we characterize how productivity shocks propagate through the spatial economy. A productivity boom in one region raises local wages and attracts migrants, bidding up housing costs and reducing fertility. But the shock also affects other regions: trading partners benefit from lower import prices, while labor market competitors lose workers and see rents decline. These cross-place spillovers mean that local policies have national consequences. Third, we show that endogenous fertility acts as a stabilizing force in spatial dynamics. Regions receiving population inflows experience rising rents and declining fertility, providing an equilibrating mechanism that moderates population growth beyond migration alone.

We calibrate the model to match data on wages, population, fertility, housing rents, migration flows, and trade flows across 741 U.S. commuting zones. The calibration recovers the key preference parameters governing substitution between consumption, housing, and children. We estimate an elasticity of substitution between housing and fertility of approximately 0.27, indicating strong complementarity. This estimate implies that policies reducing housing costs will raise fertility substantially, while policies that attract workers to expensive cities may reduce aggregate fertility through compositional effects.

The calibrated model delivers a close fit to the data. When we allow the time cost of children to vary across regions, the model exactly matches observed fertility rates. Under uniform time costs, the model (i) accounts for a meaningful share of the cross-sectional variation in fertility through the endogenous responses to wages and housing costs and (ii) reproduces the negative correlations between fertility and wages, population, and housing rents documented in Figure 1. The model also matches housing rents closely, with an R^2 of 0.85. These results suggest that the model captures the key forces shaping the spatial distribution of fertility and housing costs.

We use the calibrated model to evaluate housing supply reform. Our counterfactual equalizes inverse housing supply elasticities at the median across commuting zones, effectively relaxing constraints in large, supply-inelastic cities. The results are striking: the aggregate population growth rate rises from 0.92 to 1.17—a shift from population decline to population growth. This 27 percent increase in the growth rate demonstrates that housing supply constraints have substantial demographic consequences. The policy raises fertility disproportionately in large commuting zones, where the relaxation of supply constraints is greatest. Unlike policies that attract households into low-fertility environments, housing supply reform raises fertility precisely where population is growing, so the intensive and extensive margins reinforce rather than offset each other. The policy also reduces dispersion in real wages across space while increasing dispersion in real housing rents, reflecting the differential impact of supply constraint relaxation across locations.

These findings complement recent work on housing supply and spatial misallocation. Hsieh and Moretti (2019) show that housing supply constraints in productive cities reduce aggregate output by preventing efficient labor reallocation. Our results demonstrate that the same constraints also depress aggregate fertility, adding a demographic dimension to the welfare costs of housing regulation. The analysis suggests that housing supply reform may be a powerful lever for addressing both allocative inefficiency and demographic decline simultaneously.

Related Literature

Our paper contributes to several strands of the literature. First, we build on the quantitative spatial economics literature that studies the distribution of economic activity across space (Redding and Rossi-Hansberg, 2017; Allen and Arkolakis, 2014; Caliendo, Parro, Rossi-Hansberg, and Sarte, 2018). This literature has examined how trade costs, migration frictions, and local amenities shape the spatial equilibrium, with applications to regional policy (Fajgelbaum, Morales, Suárez Serrato, and Zidar, 2019; Fajgelbaum and Gaubert, 2020), infrastructure investment (Allen and Arkolakis, 2014), and the gains from labor mobility (Bryan and Morten, 2019a; Tombe and Zhu, 2019). A key finding is that spatial misallocation reduces aggregate output substantially (Hsieh and Moretti, 2019). Our contribution is to incorporate endogenous fertility into this framework, showing that the demographic consequences of spatial reallocation can be quantitatively important and may partially offset the efficiency gains from labor mobility.

Second, we contribute to the literature on the economics of fertility. The foundational work of Becker (1960) and Becker and Lewis (1973) established that fertility responds to the opportunity cost of children, with the quantity-quality tradeoff playing a central role. Barro and Becker (1989) embedded this choice into a dynastic growth model. More recent work has examined how housing costs affect fertility decisions (Dettling and Kearney, 2014), how labor market conditions shape family formation (Autor, Dorn, and Hanson, 2019), and how family policies influence demographic outcomes (Zhou, 2025). A growing reduced-form literature documents that house-price increases raise fertility for homeowners through wealth effects, while effects for renters are small or negative (Dettling and Kearney, 2014; Lovenheim and Mumford, 2013). Our contribution is to embed these mechanisms in a spatial general equilibrium model where housing prices respond endogenously to sorting. The observed negative correlation between prices and fertility thus conflates a compositional effect—low-fertility households sorting into expensive locations—with any behavioral response to prices. Our framework separates these channels.

Third, we connect to a growing literature on the interaction between economic geography and demography. Sato and Yamamoto (2005) and Sato (2007) develop theoretical models linking urbanization, migration, and fertility. Goto and Minamimura (2019) analyze the interaction between agglomeration economies and endogenous fertility in a two-region model. Most closely related is the concurrent paper by Arkolakis, Kwak, Woo, and Yang (2026), who develop a dynamic spatial equilibrium model with endogenous fertility calibrated to South Korea. Their paper emphasizes the quantity-quality tradeoff and education competition as drivers of low fertility in dense urban areas, showing that high-skilled parents sort into Seoul for higher wages, which intensifies education competition and raises the marginal cost of child quality. Our paper complements theirs in several respects. First, while Arkolakis et al. (2026) model fertility as a discrete choice with quality-quantity tradeoffs shaped by local education competition, we adopt a continuous fertility choice within a nested CES framework emphasizing the complementarity between housing and children. This specification yields closed-form expressions for fertility elasticities and transparently characterizes how the degree of substitutability shapes policy responses. Second, our papers target different empirical settings: they exploit rich Korean administrative data to study an extreme low-fertility environment, while we calibrate to U.S. commuting zones where fertility variation is less extreme but spatial misallocation concerns are prominent. Third, we emphasize different policy margins: while Arkolakis et al. (2026) focus on education cost reduction as the most powerful lever—finding that a nationwide reduction equivalent to 20 percent of consumption raises fertility by 23 percent—we examine housing supply reforms, finding that equalizing supply elasticities raises the population growth rate by 27 percent. Both papers reach a common conclusion: place-based policies can have large effects on aggregate fertility when they address the structural sources of low fertility in productive locations.

Fourth, our paper relates to the literature on housing supply and its macroeconomic consequences. Saiz (2010) provides estimates of housing supply elasticities across U.S. metropolitan areas. Hsieh and Moretti (2019) show that housing supply constraints in productive cities reduce aggregate output by preventing workers from accessing high-productivity locations. We extend this analysis by showing that housing constraints also affect fertility: supply-constrained cities have higher rents, which reduces family formation. This demographic channel provides an additional margin through which housing policy affects aggregate outcomes, one that operates over longer horizons than the static efficiency effects emphasized in the existing literature.

Finally, our work contributes to the literature on place-based policies. Fajgelbaum et al. (2019) develop a framework for evaluating state tax policies, showing that tax competition leads to spatial misallocation. Fajgelbaum and Gaubert (2020) characterize optimal spatial policies in the presence of agglomeration externalities. Our contribution is to introduce fertility as a policy-relevant outcome. Place-based policies have demographic consequences that should be considered alongside their effects on efficiency and equity. Our counterfactual analysis shows that housing supply reform can simultaneously improve allocative efficiency and raise aggregate fertility, suggesting that the efficiency-demography tradeoff need not bind for well-designed policies.

The remainder of the paper proceeds as follows. Section 2 presents the model and characterizes the equilibrium. Section 3 develops analytical results on the mechanisms through which local conditions affect fertility and how these effects propagate through the spatial economy. Section 4 describes the calibration strategy and data. Section 5 presents counterfactual policy exercises. Section 6 concludes.

2 Model

We develop an overlapping generations model of spatial equilibrium with endogenous fertility, migration, and housing markets. Our framework builds on the quantitative spatial economics literature (Redding and Rossi-Hansberg, 2017), extending it to incorporate demographic dynamics. The model captures how local economic conditions—particularly housing costs and wages—shape fertility decisions, and how these demographic responses feed back into the spatial distribution of economic activity.

2.1 Environment

Demographics and Geography. Time is discrete and indexed by $t = 0, 1, 2, \ldots$. The economy consists of N regions. Each agent lives for two periods: young and adult. During the young period, an agent resides with her parent in birth region o and makes no economic decisions. At the beginning of the adult period, she observes idiosyncratic preference shocks for all locations and chooses whether to remain in her birth region or migrate to another region i. In her destination, she supplies labor, consumes goods and housing, and chooses how many children to have. At the end of the adult period, she dies, and her children enter adulthood in the following period. This two-period OLG structure follows Samuelson (1958) and provides a tractable framework for studying intergenerational population dynamics.

Preferences. We assume that young agents do not consume. In period t, an adult agent ν who moves from region o to region i has a probability of ζ_i to be a homeowner. We denote $j \in \{O,R\}$ as the status index where O refers to homeowner and R refers to renter. The agent ν moving from o to i with status j has the utility

$$U^j_{oi,t}(\nu) = \frac{B_{i,t}(\nu)}{d_{oi,t}} V^j_{i,t}(\nu), \quad j \in \{O,R\},$$
$$V^j_{i,t}(\nu) \equiv \left[\alpha^{\frac{1}{\gamma}} C^j_{i,t}(\nu)^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h^j_{i,t}(\nu)^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n^j_{i,t}(\nu)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}, \tag{1}$$

where $C^j_{i,t}(\nu)$ denotes consumption of a composite good, $h^j_{i,t}(\nu)$ denotes housing services, and $n^j_{i,t}(\nu)$ denotes the number of children. The term $d_{oi,t} \geq 1$ represents the iceberg migration friction between regions o and i, with $d_{oo,t} = 1$ for stayers.

The preference specification embeds a nested CES structure that governs substitution patterns across the three arguments of the utility function. The outer nest, governed by parameter $\gamma \geq 0$, determines the elasticity of substitution between consumption goods and a composite of housing and children. The inner nest, governed by $\rho \geq 0$, determines the elasticity of substitution between housing and children. The parameters $\alpha \in (0,1)$ and $\beta \in (0,1)$ control expenditure shares. This nested structure allows for flexible substitution patterns: when housing becomes expensive, households may substitute toward fewer children (if $\rho > 1$) or toward consumption goods (if $\gamma > 1$). The interaction between these margins is central to our analysis of how housing costs affect fertility.

The term $B_{i,t}(\nu)$ captures idiosyncratic taste shocks for location i, which generate dispersion in location choices even among otherwise identical agents. Following McFadden (1973) and the subsequent spatial economics literature (Eaton and Kortum, 2002; Bryan and Morten, 2019b; Redding, 2016), we assume these shocks are drawn from a Fréchet distribution:

$$\Pr\left[B_{i,t}(\nu) \leq b\right] = \exp\left[-B_{i,t} b^{-\eta}\right], \tag{2}$$

where $B_{i,t} > 0$ is a location-specific amenity shifter and $\eta > 1$ governs the dispersion of idiosyncratic tastes. A higher η implies less dispersion, making agents more responsive to systematic differences in utility across locations.

Production. Each region produces a distinct variety of goods under perfect competition. Varieties are aggregated through a CES consumption index with elasticity of substitution $\sigma > 1$, following Dixit and Stiglitz (1977) and Krugman (1980). The unit cost of producing variety i is

$$uc_{i,t} = \frac{w_{i,t}}{A_{i,t}\left(L^e_{i,t}\right)^{\psi}}, \tag{3}$$

where $w_{i,t}$ denotes the wage, $A_{i,t}$ is an exogenous productivity shifter, $L^e_{i,t}$ is effective labor in region i, and $\psi \geq 0$ captures external economies of scale. The agglomeration externality reflects productivity spillovers from the concentration of economic activity, as emphasized by Ciccone and Hall (1996) and Combes, Duranton, Gobillon, Puga, and Roux (2012). All varieties are tradable, subject to iceberg trade costs $\tau_{in,t} \geq 1$ from region i to region n, with $\tau_{ii,t} = 1$.

Housing Supply. Following Saiz (2010) and Hsieh and Moretti (2019), the inverse supply function of housing in region i is

$$r_{i,t} = \bar{r}_{i,t} H^{\xi_i}_{i,t}, \tag{4}$$

where $\xi_i \geq 0$ is the inverse elasticity of housing supply, $H_{i,t}$ is total housing supply, and $\bar{r}_{i,t}$ is an exogenous shifter. A higher ξ_i implies more inelastic supply: as housing demand increases, rents rise more sharply in supply-constrained regions.

Household Problem. The initial distribution of adults, $L_{i,0}$, is given exogenously. Each adult is endowed with one unit of time, which she allocates between market work and parenting. Raising each child requires $\chi_{i,t}$ units of time, capturing the opportunity cost of fertility in terms of forgone labor income. This formulation follows Becker (1960) and Becker and Lewis (1973).

Conditional on moving to region i from region o in period t, the adult solves:

$$V_{i,t}^{j} = \max_{\left\{C_{i,t}^{j}, h_{i,t}^{j}, n_{i,t}^{j}\right\}} \left[\alpha^{\frac{1}{\gamma}} \left(C_{i,t}^{j}\right)^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} \left(h_{i,t}^{j}\right)^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} \left(n_{i,t}^{j}\right)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}} \tag{5}$$
$$\text{s.t. } P_{i,t} C_{i,t}^{j} + r_{i,t} h_{i,t}^{j} = w_{i,t} \left(1 - \chi_{i,t} n_{i,t}^{j}\right) + T_{i,t}^{j},$$

where $P_{i,t}$ is the price index of the consumption good composite, $r_{i,t}$ is the housing rent, and $T_{i,t}$ is a lump-sum transfer. The right-hand side of the budget constraint reflects the opportunity cost structure of fertility: each child reduces labor income by $\chi_{i,t} w_{i,t}$.

Housing Revenue Distribution. Housing revenues accrue to homeowners, distributed as follows:

$$\begin{aligned} T_{i,t}^{O} &= \frac{1}{\zeta_i} \left[\frac{1}{2} \frac{r_{i,t} H_{i,t}}{L_{i,t}} + \frac{\sum_{k=1}^{N} \frac{1}{2} r_{k,t} H_{k,t}}{\sum_{k=1}^{N} L_{k,t}} \right], \\ T_{i,t}^{R} &= 0, \\ T_{i,t} &\equiv \zeta_i T_{i,t}^{O} + (1-\zeta_i) T_{i,t}^{R} = \frac{1}{2} \frac{r_{i,t} H_{i,t}}{L_{i,t}} + \frac{\sum_{k=1}^{N} \frac{1}{2} r_{k,t} H_{k,t}}{\sum_{k=1}^{N} L_{k,t}}. \end{aligned} \tag{6}$$

2.2 Equilibrium

We now characterize the equilibrium. Let $L_{oi,t}$ denote the number of adults born in region o who live in region i in period t, and let $L_{i,t}$ denote total adult population:

$$L_{i,t} \equiv \sum_{o=1}^{N} L_{oi,t}. \tag{7}$$

Effective labor supply equals population minus time devoted to childrearing:

$$L_{i,t}^{e} = (1 - \chi_{i,t} n_{i,t}) L_{i,t}, \quad n_{i,t} \equiv \zeta_i n_{i,t}^{O} + (1-\zeta_i) n_{i,t}^{R}. \tag{8}$$

This expression highlights a key margin: higher fertility reduces effective labor supply, affecting local productivity through agglomeration and, consequently, wages and prices throughout the economy.

Optimal Consumption and Fertility. The household's optimization yields demand functions for goods, housing, and children. Define the expenditure index:

$$\Xi_{i,t} \equiv \alpha P_{i,t}^{1-\gamma} + (1-\alpha)\left[\beta r_{i,t}^{1-\rho} + (1-\beta)\left(\chi_{i,t} w_{i,t}\right)^{1-\rho}\right]^{\frac{1-\gamma}{1-\rho}}. \tag{9}$$

Optimal goods consumption, $C_{i,t} \equiv \zeta_i C_{i,t}^{O} + (1-\zeta_i)\, C_{i,t}^{R}$, is:

$$C_{i,t} = \frac{1}{P_{i,t}} \frac{\alpha P_{i,t}^{1-\gamma}}{\Xi_{i,t}} \left(w_{i,t} + T_{i,t}\right). \tag{10}$$

Optimal housing consumption, $h_{i,t} \equiv \zeta_i h_{i,t}^{O} + (1-\zeta_i)\, h_{i,t}^{R}$, is:

$$h_{i,t} = \frac{1}{r_{i,t}} \frac{\beta r_{i,t}^{1-\rho}}{\beta r_{i,t}^{1-\rho} + (1-\beta)\left(\chi_{i,t} w_{i,t}\right)^{1-\rho}} \frac{(1-\alpha)\left[\beta r_{i,t}^{1-\rho} + (1-\beta)\left(\chi_{i,t} w_{i,t}\right)^{1-\rho}\right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_{i,t}} \left(w_{i,t} + T_{i,t}\right). \tag{11}$$

Optimal fertility, $n_{i,t} \equiv \zeta_i n_{i,t}^{O} + (1-\zeta_i)\, n_{i,t}^{R}$, is:

$$n_{i,t} = \frac{1}{\chi_{i,t} w_{i,t}} \frac{(1-\beta)\left(\chi_{i,t} w_{i,t}\right)^{1-\rho}}{\beta r_{i,t}^{1-\rho} + (1-\beta)\left(\chi_{i,t} w_{i,t}\right)^{1-\rho}} \frac{(1-\alpha)\left[\beta r_{i,t}^{1-\rho} + (1-\beta)\left(\chi_{i,t} w_{i,t}\right)^{1-\rho}\right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_{i,t}} \left(w_{i,t} + T_{i,t}\right). \tag{12}$$

These expressions reveal the key economic forces shaping fertility. The fertility rate depends on the ratio of housing rents to the opportunity cost of children, $r_{i,t}/(\chi_{i,t} w_{i,t})$, through the inner CES aggregator. When housing is expensive relative to the time cost of children, households substitute toward fewer children and more housing (if $\rho > 1$). Income raises fertility through an income effect, while higher wages also raise the opportunity cost of children, generating a substitution effect working in the opposite direction.

Trade and Prices. Goods trade follows the standard gravity structure. The share of region n's expenditure on goods from region i is:

$$\lambda_{in,t} \equiv \frac{X_{in,t}}{X_{n,t}} = \left(\tau_{in,t} w_{i,t}\right)^{1-\sigma} \left(A_{i,t}\left(L_{i,t}^{e}\right)^{\psi}\right)^{\sigma-1} P_{n,t}^{\sigma-1}, \tag{13}$$

where the price index in region n is:

$$P_{n,t} = \left[\sum_{i=1}^{N} \left(\tau_{in,t} w_{i,t} \right)^{1-\sigma} \left(A_{i,t} \left(L_{i,t}^{e} \right)^{\psi} \right)^{\sigma-1} \right]^{\frac{1}{1-\sigma}}. \tag{14}$$

Market Clearing. Wages are determined by labor market clearing:

$$w_{i,t} L_{i,t}^{e} = \sum_{n=1}^{N} \lambda_{in,t} X_{n,t}. \tag{15}$$

Total goods expenditure equals consumption expenditure:

$$X_{i,t} = \frac{\alpha P_{i,t}^{1-\gamma}}{\Xi_{i,t}} \left(w_{i,t} + T_{i,t} \right) L_{i,t}. \tag{16}$$

Housing market clearing requires:

$$H_{i,t} = \frac{L_{i,t}}{r_{i,t}} \left[\frac{\beta r_{i,t}^{1-\rho}}{\beta r_{i,t}^{1-\rho} + (1-\beta) \left(\chi_{i,t} w_{i,t} \right)^{1-\rho}} \frac{(1-\alpha) \left[\beta r_{i,t}^{1-\rho} + (1-\beta) \left(\chi_{i,t} w_{i,t} \right)^{1-\rho} \right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_{i,t}} \left(w_{i,t} + T_{i,t} \right) \right]. \tag{17}$$

Migration and Population Dynamics. The number of young agents in region o at the end of period $t-1$ equals $n_{o,t-1} L_{o,t-1}$. These agents become adults in period t and choose where to live. Given the Fréchet distribution, the share of adults born in o who migrate to i is:

$$\frac{L_{oi,t}}{n_{o,t-1} L_{o,t-1}} = \frac{B_{i,t} \left(\frac{1}{d_{oi,t}} V_{i,t} \right)^{\eta}}{\sum_{k=1}^{N} B_{k,t} \left(\frac{1}{d_{ok,t}} V_{k,t} \right)^{\eta}}, \tag{18}$$

where indirect utility is:

$$V_{i,t} = \left[\alpha^{\frac{1}{\gamma}} C_{i,t}^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h_{i,t}^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n_{i,t}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1} \frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}. \tag{19}$$

This migration equation embodies the core spatial equilibrium logic: regions with higher utility attract more migrants. The presence of fertility in both the numerator of (18)—through the origin region's fertility rate $n_{o,t-1}$—and in the destination's attractiveness $V_{i,t}$ creates a rich dynamic interaction between demographic and spatial forces.

Equilibrium Definition.

Definition 1 (Equilibrium)**.** *Given exogenous fundamentals* $(A_{i,t}, B_{i,t}, d_{oi,t}, \tau_{in,t}, \chi_{i,t}, \bar{r}_{i,t})$*, parameters* $(\eta, \psi, \sigma, \gamma, \rho, \alpha, \beta, \xi_i)$*, and initial conditions* $(L_{i,0})$*, a dynamic equilibrium is a sequence of prices* $(w_{i,t}, r_{i,t}, P_{i,t})$*, allocations* $(n_{i,t}, H_{i,t})$*, and population distributions* $(L_{i,t})$ *such that for all* $t \geq 0$*:*

1. *Wages* $w_{i,t}$ *clear the labor market according to Equation* (15).
2. *Housing rents* $r_{i,t}$ *clear the housing market according to Equation* (17).
3. *Housing supply* $H_{i,t}$ *satisfies Equation* (4).
4. *Price indices* $P_{i,t}$ *are given by Equation* (14).
5. *Fertility rates* $n_{i,t}$ *satisfy Equation* (12).
6. *Population* $L_{i,t}$ *evolves according to Equations* (18) *and* (7).

Fertility Rate of Homeowner and Renter. Given homeowner share ζ_i and equilibrium outcomes $(n_{i,t}, H_{i,t}, w_{i,t}, r_{i,t})$, the fertility rates $\left(n_{i,t}^O, n_{i,t}^R\right)$ can be solved by

$$\begin{aligned} &\zeta_i n_{i,t}^O + (1-\zeta_i)\, n_{i,t}^R = n_{i,t} \\ &\frac{n_{i,t}^O}{n_{i,t}^R} = \frac{1}{w_{i,t}}\left[w_{i,t} + \frac{1}{\zeta_i}\left(\frac{1}{2}\frac{r_{i,t}H_{i,t}}{L_{i,t}} + \frac{\sum_{k=1}^N \frac{1}{2} r_{k,t} H_{k,t}}{\sum_{k=1}^N L_{k,t}}\right)\right] \end{aligned} \tag{20}$$

Expected Welfare. The Fréchet structure yields a convenient expression for expected welfare. The expected utility of an agent born in region o who enters adulthood in period t is:

$$W_{o,t} = \Gamma\left(1 - \frac{1}{\eta}\right)\left[\sum_{k=1}^N B_{k,t}\left(\frac{1}{d_{ok,t}} V_{k,t}\right)^{\eta}\right]^{\frac{1}{\eta}}, \tag{21}$$

where $\Gamma(\cdot)$ is the gamma function. This welfare measure aggregates opportunities across all potential destinations and forms the basis for our welfare analysis.

2.3 Steady State

To characterize long-run properties, we analyze steady states in which population shares remain constant over time.

Definition 2 (Steady State)**.** *A steady state is an equilibrium in which population in all regions grows at a common constant rate:*

$$L_{i,t} = gL_{i,t-1}, \quad g > 0, \quad \forall i, t. \tag{22}$$

In steady state, the growth rate g equals the population-weighted average fertility rate:

$$g = \frac{\sum_{i=1}^{N} n_{i,t} L_{i,t}}{\sum_{i=1}^{N} L_{i,t}}, \quad n_{i,t} \equiv \zeta_i n_{i,t}^{O} + (1-\zeta_i)\, n_{i,t}^{R}, \quad \forall t. \tag{23}$$

This condition ensures demographic consistency: the number of children born each period equals the number of new adults. If $g > 1$, the population grows; if $g < 1$, it declines. The steady-state growth rate is determined endogenously by the interaction of local fertility decisions, migration patterns, and economic conditions.

We drop time subscripts when expressing steady-state quantities and denote detrended population as $L_i \equiv L_{i,t} g^{-t}$.

Definition 3 (Steady-State Equilibrium)**.** *Given fundamentals* $(A_i, B_i, d_{oi}, \tau_{in}, \chi_i, \bar{r}_i)$ *and parameters* $(\eta, \psi, \sigma, \gamma, \rho, \alpha, \beta, \xi_i)$*, a steady-state equilibrium consists of prices* (w_i, r_i, P_i)*, allocations* (n_i, H_i)*, population* (L_i)*, and growth rate* g *such that:*

1. ***Labor market clearing:*** *Wages* (w_i) *satisfy*

$$\begin{aligned} w_i L_i^e &= \sum_{n=1}^{N} \lambda_{in} P_n C_n L_n, \quad \lambda_{in} = (\tau_{in} w_i)^{1-\sigma} \left(A_i \left(L_i^e\right)^{\psi}\right)^{\sigma-1} P_n^{\sigma-1} \\ L_i^e &= (1-\chi_i n_i)\, L_i, \quad C_n = \frac{1}{P_n} \frac{\alpha P_n^{1-\gamma}}{\Xi_n} \left(w_n + T_n\right), \end{aligned} \tag{24}$$

where the transfer is

$$T_n = \frac{1}{2}\frac{r_n H_n}{L_n} + \frac{\sum_{k=1}^{N} \frac{1}{2} r_k H_k}{\sum_{k=1}^{N} L_k}, \tag{25}$$

and the expenditure index is

$$\Xi_i \equiv \alpha P_i^{1-\gamma} + (1-\alpha)\left[\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}\right]^{\frac{1-\gamma}{1-\rho}}. \tag{26}$$

2. ***Housing market clearing:*** *Rents* (r_i) *and housing supply* (H_i) *satisfy*

$$r_i = \bar{r}_i H_i^{\xi_i}, \quad H_i = h_i L_i, \quad h_i = \frac{1}{r_i}\left[\frac{\beta r_i^{1-\rho}}{\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}} \frac{(1-\alpha)\left[\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}\right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_i}(w_i + T_i)\right]. \tag{27}$$

3. ***Price indices:*** (P_i) *satisfy*

$$P_n = \left[\sum_{i=1}^{N} (\tau_{in} w_i)^{1-\sigma} \left(A_i (L_i^e)^{\psi}\right)^{\sigma-1}\right]^{\frac{1}{1-\sigma}}. \tag{28}$$

4. ***Fertility:*** *Fertility rates* (n_i) *satisfy*

$$n_i = \frac{1}{\chi_i w_i} \frac{(1-\beta)(\chi_i w_i)^{1-\rho}}{\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}} \frac{(1-\alpha)\left[\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}\right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_i}(w_i + T_i). \tag{29}$$

5. ***Population distribution:*** *The detrended population* (L_i) *satisfies*

$$gL_i = \sum_{o=1}^{N} \frac{B_i\left(\frac{1}{d_{oi}} V_i\right)^{\eta}}{\sum_{k=1}^{N} B_k\left(\frac{1}{d_{ok}} V_k\right)^{\eta}} n_o L_o, \tag{30}$$

where $V_i = \left[\alpha^{\frac{1}{\gamma}} C_i^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}}\left[\beta^{\frac{1}{\rho}} h_i^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n_i^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}\frac{\gamma-1}{\gamma}}\right]^{\frac{\gamma}{\gamma-1}}.$

6. ***Aggregate fertility:*** *The growth rate g satisfies*

$$g = \frac{\sum_{i=1}^{N} n_i L_i}{\sum_{i=1}^{N} L_i}. \tag{31}$$

Several features of this equilibrium merit emphasis. First, the population growth rate g is endogenous and depends on the distribution of economic activity across regions with different fertility rates. Policies that reallocate population toward high-fertility regions raise aggregate fertility and the long-run growth rate. Second, the interaction between housing supply elasticities and fertility creates spatial variation in demographic responses to economic shocks. Third, migration links regions together: a shock to one region affects fertility and welfare throughout the economy through trade, migration, and price adjustments.

3 Model Mechanisms

This section develops analytical results to illuminate how the model's key mechanisms operate. We derive first-order approximations around the steady state to characterize how local shocks propagate through the spatial economy. Throughout, we highlight the role of endogenous fertility in shaping both local and aggregate responses.

3.1 Local Determinants of Fertility

We begin by characterizing how fertility responds to local economic conditions. Define the expenditure shares on consumption goods, housing, and children as:

$$\omega_i^C \equiv \frac{P_i C_i}{w_i + T_i}, \quad \omega_i^h \equiv \frac{r_i h_i}{w_i + T_i}, \quad \omega_i^n \equiv \frac{\chi_i w_i n_i}{w_i + T_i}, \tag{32}$$

and the within-nest shares as:

$$\tilde{\omega}_i^h \equiv \frac{\beta r_i^{1-\rho}}{\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}}, \quad \tilde{\omega}_i^n \equiv 1 - \tilde{\omega}_i^h. \tag{33}$$

A key property of the nested CES structure is that optimal fertility, housing consumption, and goods consumption are all linear in total income $w_i + T_i^j$ for a household of tenure type $j \in \{O, R\}$. This linearity implies that aggregate local fertility $n_i = \zeta_i n_i^O + (1-\zeta_i) n_i^R$ depends only on the average transfer $T_i = \zeta_i T_i^O + (1-\zeta_i) T_i^R$, not on how housing revenues are distributed between owners and renters. Ownership status redistributes fertility from renters to owners—since $n_i^O > n_i^R$ when $T_i^O > 0 = T_i^R$—but leaves aggregate local fertility unchanged. We therefore characterize mechanisms in terms of aggregate local fertility without loss of generality.

Proposition 1 (Local Fertility Elasticities)**.** *Holding transfers T_i fixed, the elasticities of aggregate local fertility with respect to wages, rents, and the time cost of children are:*

$$\varepsilon_{n,w} \equiv \left.\frac{\partial \log n_i}{\partial \log w_i}\right|_{T_i} = \underbrace{-\rho - (\gamma - \rho)\tilde{\omega}_i^n - (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n}_{\text{substitution effect}} + \underbrace{\frac{w_i}{w_i + T_i}}_{\text{income effect}} \tag{34}$$

$$\varepsilon_{n,r} \equiv \left.\frac{\partial \log n_i}{\partial \log r_i}\right|_{T_i} = \underbrace{\tilde{\omega}_i^h \left[-(\gamma - \rho) - (1-\gamma)(1-\omega_i^C)\right]}_{\text{price effect}} \tag{35}$$

$$\varepsilon_{n,\chi} \equiv \frac{\partial \log n_i}{\partial \log \chi_i} = \varepsilon_{n,w} - \frac{w_i}{w_i + T_i} \tag{36}$$

The price effect $\varepsilon_{n,r}$ is negative if and only if $-(1-\gamma)(1-\omega_i^C) < \gamma - \rho$. This condition holds whenever $\rho \leq \gamma < 1$. When transfers respond endogenously to rents, the total rent elasticity includes an additional wealth effect:

$$\frac{\partial \log n_i}{\partial \log r_i} = \varepsilon_{n,r} + \frac{T_i}{w_i + T_i} \cdot \varepsilon_{T,r}, \tag{37}$$

where $\varepsilon_{T,r} \equiv \partial \log T_i / \partial \log r_i > 0$.

Proof. See Appendix A.1. □

These elasticities admit clear economic interpretations. Consider first the wage elasticity $\varepsilon_{n,w}$. When wages rise, the opportunity cost of children $\chi_i w_i$ increases, making children relatively more expensive. This generates a *substitution effect* with three components:

1. The direct effect $-\rho$ captures the own-price elasticity of fertility demand within the inner nest, holding composite prices fixed.

2. The within-nest feedback $-(\gamma-\rho)\tilde{\omega}_i^n$ reflects how the higher child cost raises the housing-fertility composite price, which in turn affects fertility demand. When $\gamma > \rho$, the outer nest is more substitutable than the inner nest, amplifying the response; when $\gamma < \rho$, it dampens the response.

3. The cross-nest substitution $-(1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n$ captures reallocation toward consumption goods. When $\gamma < 1$, consumption and the housing-fertility composite are complements, so the higher composite price reduces demand for both.

Working against the substitution effect is the *income effect*: higher wages expand the budget set, and since children are a normal good, fertility rises with income. The income effect is captured by $w_i/(w_i+T_i)$, the share of income derived from wages. The net wage elasticity is negative when the substitution effect dominates.

The rent elasticity $\varepsilon_{n,r}$ captures how housing costs affect fertility through the *price effect*. Higher rents raise the price of housing within the inner nest and thereby increase the price index of the housing-fertility composite. The structure parallels the wage elasticity but without the direct $-\rho$ term, since rents do not directly enter the opportunity cost of children. The overall effect is negative when $\rho \leq \gamma < 1$.

The time cost elasticity $\varepsilon_{n,\chi}$ equals the substitution component of the wage elasticity. The relationship $\varepsilon_{n,\chi} = \varepsilon_{n,w} - w_i/(w_i+T_i)$ follows because χ_i and w_i enter the opportunity cost multiplicatively: both raise the effective price $\chi_i w_i$, but only wages generate an income effect. This relationship implies that the time cost elasticity is always more negative than the wage elasticity.

Corollary 1 (Local Impacts of Rents on Fertility Rates of Homeowners and Renters)**.** *The partial-equilibrium effects of rents on the steady-state fertility rates of homeowners and renters can be expressed as:*

$$\begin{aligned}
\varepsilon_{n^O,r} &\equiv \frac{\partial \log n_i^O}{\partial \log r_i} = \varepsilon_{n,r} + \frac{T_i}{w_i+T_i}\varepsilon_{T,r} + \left(1-\tilde{\zeta}_i\right)\frac{\frac{1}{\zeta_i}T_i}{w_i+\frac{1}{\zeta_i}T_i}\varepsilon_{T,r} \\
\varepsilon_{n^R,r} &\equiv \frac{\partial \log n_i^R}{\partial \log r_i} = \varepsilon_{n,r} + \frac{T_i}{w_i+T_i}\varepsilon_{T,r} - \tilde{\zeta}_i\frac{\frac{1}{\zeta_i}T_i}{w_i+\frac{1}{\zeta_i}T_i}\varepsilon_{T,r}
\end{aligned}, \tag{38}$$

where $\tilde{\zeta}_i \equiv \zeta_i \frac{n_i^O}{n_i}$*. Therefore, we have*

$$\frac{\partial \log\left(n_i^O / \log n_i^R\right)}{\partial \log r_i} = \varepsilon_{n^O,r} - \varepsilon_{n^R,r} = \frac{\frac{1}{\zeta_i} T_i}{w_i + \frac{1}{\zeta_i} T_i} \varepsilon_{T,r} > 0. \tag{39}$$

Proof. See Appendix A.1. □

Since housing revenues are rebated to homeowners, their local fertility elasticities incorporate an additional income effect, $\left(1 - \tilde{\zeta} i\right) \frac{\frac{1}{\zeta_i} T_i}{w_i + \frac{1}{\zeta_i} T_i} \varepsilon_{T,r}$. Consequently, an increase in local housing rents raises the ratio of fertility rates between homeowners and renters, consistent with the empirical evidence in Dettling and Kearney (2014).

3.2 General Equilibrium Propagation

Local shocks propagate through the spatial economy via goods trade, worker migration, and price adjustment. Understanding these channels is essential for policy analysis, since interventions in one region affect outcomes elsewhere.

Proposition 2 (Spatial Propagation of Shocks)**.** *Consider a productivity shock* $\hat{A}_j > 0$ *in region* j*, where* $\hat{x} \equiv d\log x$ *denotes log-deviations. The shock propagates through three channels:*

Trade channel. *Holding wages and employment fixed, region* i*'s price index responds to productivity in region* j *according to:*

$$\left.\frac{\partial \log P_i}{\partial \log A_j}\right|_{w,L^e} = -\lambda_{ij} < 0, \tag{40}$$

where λ_{ij} *is region* i*'s expenditure share on goods from* j*. In general equilibrium, wages and employment adjust, yielding:*

$$\hat{P}_i = \sum_k \lambda_{ik} \left(\hat{w}_k - \hat{A}_k - \psi \hat{L}_k^e\right). \tag{41}$$

Migration channel. *In steady state, population satisfies* $L_i = \sum_o m_{oi} n_o L_o$*, where* $m_{oi} = B_i (d_{oi}^{-1} V_i)^\eta / \Phi_o$ *is the migration probability and* $\Phi_o \equiv \sum_k B_k (d_{ok}^{-1} V_k)^\eta$*. Define the expected utility change for agents* *born in region o as $\bar{\hat{V}}_o \equiv \sum_k m_{ok}\hat{V}_k$. Log-linearizing around steady state:*

$$\hat{L}_i = \sum_o \pi_{oi}\left[\eta(\hat{V}_i - \bar{\hat{V}}_o) + \hat{n}_o + \hat{L}_o\right], \tag{42}$$

where $\pi_{oi} \equiv m_{oi}n_oL_o/L_i$ is the share of region i's population originating from region o.

Housing price channel. *Population and per-capita housing demand changes affect rents through the housing supply curve:*

$$\hat{r}_i = \xi_i(\hat{h}_i + \hat{L}_i), \tag{43}$$

where ξ_i is the inverse housing supply elasticity and housing market clearing requires total housing $H_i = h_iL_i$.

Proof. See Appendix A.2. □

These three channels interact to determine the spatial propagation of shocks. In the region receiving a positive productivity shock, wages rise and migrants arrive, bidding up housing costs. Both higher wages and higher rents reduce fertility—wages through the substitution effect, rents through the price effect—generating a fertility decline in booming regions.

In other regions, the effects are more nuanced. Trading partners benefit from lower import prices through equation (40), which raises real income and may increase fertility. Regions losing migrants experience reduced housing demand. Through equation (43), this translates into lower rents, which raises fertility through the price effect. The net effect on aggregate fertility depends on whether these offsetting forces balance.

A key implication is that endogenous fertility acts as a stabilizing force in spatial dynamics.

Proposition 3 (Fertility as a Stabilizing Force)**.** *Consider a positive productivity shock in region j that raises local wages and rents, thereby reducing local fertility: $\hat{n}_j < 0$. Comparing steady states—defined as equilibria where population shares are constant—before and after the shock:*

$$\hat{L}_j^{SS}\big|_{endogenous\ n} < \hat{L}_j^{SS}\big|_{exogenous\ n}. \tag{44}$$

That is, endogenous fertility dampens the long-run population response to productivity shocks.

Proof. See Appendix A.3. □

This result helps explain why highly productive cities have not grown even faster despite sustained migration inflows. The migration-induced increase in housing costs depresses fertility, partially offsetting population growth through natural increase. Endogenous fertility thus provides an equilibrating mechanism that moderates spatial concentration. The magnitude of this dampening depends on the natural retention rate $\mu_j \equiv m_{jj} n_j$—the fraction of population that stays and reproduces locally—with larger μ_j implying stronger stabilization.

3.3 Aggregate Fertility and the Extensive Margin

Aggregate fertility is a population-weighted average of local fertility rates. Changes in aggregate fertility therefore reflect both local fertility changes (the intensive margin) and population reallocation across locations with different fertility rates (the extensive margin).

Proposition 4 (Decomposition of Aggregate Fertility)**.** *Aggregate fertility $\bar{n} \equiv \sum_i n_i L_i / \bar{L}$, where $\bar{L} \equiv \sum_i L_i$, evolves according to:*

$$d\bar{n} = \underbrace{\sum_i s_i \, dn_i}_{\text{intensive margin}} + \underbrace{\sum_i (n_i - \bar{n}) \, ds_i}_{\text{extensive margin}}, \tag{45}$$

where $s_i \equiv L_i / \bar{L}$ is region i's population share. In terms of log changes:

$$\hat{\bar{n}} = \sum_i \frac{s_i n_i}{\bar{n}} \hat{n}_i + \frac{1}{\bar{n}} Cov_s(n_i, \hat{L}_i), \tag{46}$$

where $Cov_s(n_i, \hat{L}_i) \equiv \sum_i s_i (n_i - \bar{n}) \hat{L}_i$ is the population-weighted covariance between fertility levels and population growth rates. The extensive margin is negative when population flows toward regions with below-average fertility.

Proof. See Appendix A.4. □

The decomposition reveals a potential tension between allocative efficiency and aggregate fertility. Policies that reallocate population toward high-productivity, high-wage, expensive locations may raise aggregate output while reducing aggregate fertility through composition effects. The extensive margin $\text{Cov}_s(n_i, \hat{L}_i)/\bar{n}$ is negative precisely when high-growth regions have low fertility—the empirical pattern documented in Figure 1.

Importantly, this tradeoff is not inevitable. Policies reducing the cost of living in productive regions can raise local fertility through the price effect while simultaneously attracting migrants. When a policy raises fertility in locations that are gaining population—that is, when the intensive margin raises n_i in regions where $\hat{L}_i > 0$—the two margins reinforce rather than offset each other. This observation motivates our counterfactual analysis of housing supply reform: by reducing rents in supply-constrained productive cities, such reform raises fertility precisely where population growth is concentrated.

3.4 Summary

The analytical results yield five key insights for understanding the spatial distribution of fertility and evaluating place-based policies:

1. *Higher wages reduce fertility* when the substitution effect—comprising the direct price effect, within-nest feedback, and cross-nest substitution toward consumption—dominates the income effect. The calibrated complementarities ($\rho < 1$ and $\gamma < 1$) ensure this condition holds.

2. *Higher housing costs reduce fertility* through the price effect when $\gamma < 1$ and consumption is not too large a share of expenditure. This negative effect may be partially offset by a positive wealth effect when housing revenues are distributed to local residents.

3. *Local shocks propagate across space* through trade (affecting price indices), migration (reallocating population), and housing markets (translating population changes into rent changes). These channels generate heterogeneous fertility responses across regions even to localized shocks.

4. *Endogenous fertility stabilizes spatial dynamics* by dampening population responses to productivity shocks. Regions experiencing booms see fertility decline, partially offsetting migration-driven population growth.

5. *Aggregate fertility depends on both the intensive and extensive margins*. Policies that attract population to low-fertility locations reduce aggregate fertility through composition effects, but policies that reduce costs in productive cities can align the two margins.

4 Model Calibration

This section describes our calibration strategy. We calibrate the model to match data on wages, population, fertility, housing rents, and trade and migration flows across 741 U.S. commuting zones, drawn from the 2000 U.S. Census via IPUMS USA (Ruggles et al., 2025). The calibration proceeds in four steps. First, we assign values to parameters taken from the existing literature. Second, we estimate trade and migration costs using bilateral flow data. Third, we calibrate preference parameters using simulated method of moments. Finally, we recover region-specific fundamentals to match the observed spatial distribution exactly.

4.1 Data and Pre-Determined Parameters

Table 1 summarizes the pre-determined parameters and data sources. The agglomeration elasticity $\psi = 0.1$ lies in the middle of the range of estimates in the urban economics literature, implying that doubling local employment raises productivity by approximately 7 percent (Combes et al., 2012; Rosenthal and Strange, 2004). The migration elasticity $\eta = 1.5$ follows Tombe and Zhu (2019), implying that a 10 percent increase in real income increases the migration flow by approximately 15 percent. The trade elasticity $\sigma - 1 = 4$ is standard in the quantitative trade literature (Head and Mayer, 2014).

The inverse housing supply elasticity ξ_i uses metropolitan-area-level estimates from Saiz (2010), based on geographic constraints and land use regulations. For commuting zones without Saiz estimates, we assign the median supply elasticity. The housing supply shifter $\bar{r}_i$ is recovered to match observed rents and quantities.

4.2 Trade and Migration Costs

We estimate trade costs τ_{in} and migration costs d_{oi} using gravity regressions on bilateral flow data. Both costs take a power-law form in geographic distance.

Trade Costs. We assume $\tau_{in} = D_{in}^{\delta_\tau}$, where D_{in} is the great-circle distance between regions i and n, computed from geographic coordinates in Chetty, Stepner, Abraham, Lin, Scuderi, Turner,

Table 1: Data and Pre-Determined Parameters

Pre-Determined Parameters	
Parameter	Sources
ψ: agglomeration	$= 0.1$ from the literature
η: migration elasticity	$= 1.5$ from Tombe and Zhu (2019)
$\sigma - 1$: trade elasticity	$= 4$ from the literature
ξ_i: inverse supply elasticity of housing	Saiz (2010)
$\bar{r}_i$: housing supply shifter	housing rent and quantity
Data	
Variable	Data Source
λ_{in}: Trade flows across U.S. regions	Commodity Flow Survey and gravity estimation
L_{oi}: Migration flows across U.S. regions	Census 2000 county-to-county migration flows
n_i: Fertility rate	IPUMS USA, Census 2000
w_i: Wage	IPUMS USA, Census 2000
L_i: Population	IPUMS USA, Census 2000
h_i: Per-capita housing	IPUMS USA, Census 2000
r_i: Housing rent	IPUMS USA, Census 2000

Bergeron, and Cutler (2016). The distance elasticity δ_τ follows Ramondo, Rodríguez-Clare, and Saborío-Rodríguez (2016), who estimate

$$\log \lambda_{in} = -(1-\sigma)\,\delta_\tau \log D_{in} + \mathrm{fe}_i + \mathrm{fe}_n + \tilde{u}_{in}, \tag{47}$$

where λ_{in} is the bilateral trade share from state i to n, and the fixed effects absorb origin productivity and destination demand shifters.

Migration Costs. We assume $d_{oi} = D_{oi}^{\delta_d}$ and estimate:

$$\log L_{oi} = -\eta\,\delta_d \log D_{oi} + \mathrm{fe}_o + \mathrm{fe}_i + \tilde{\varepsilon}_{oi}, \tag{48}$$

where L_{oi} is the number of migrants from o to i. The estimated coefficient is $\widehat{\eta\,\delta_d} = -2.09$ (standard error 0.01), indicating a strongly negative relationship between distance and migration flows. Given $\eta = 1.5$, this implies $\delta_d = 1.4$. Doubling the distance reduces migration flows by approximately a factor of $2^{1.4} \approx 2.6$.

4.3 Preference Parameters

The key structural parameters are the elasticity of substitution between consumption and the housing-fertility composite (γ), the elasticity between housing and fertility (ρ), the consumption share (α), and the housing share within the housing-fertility composite (β). We calibrate these using simulated method of moments.

Identification Strategy. Table 2 summarizes the mapping between parameters and moments. The correlation between fertility and wages identifies γ: when $\gamma < 1$ (complementarity), higher wages reduce demand for both housing and children. The correlation between fertility and population identifies ρ: population affects fertility through housing costs, and the sign and magnitude of this relationship reveal the degree of complementarity between housing and children. Average fertility pins down α, and the aggregate rent-to-income ratio identifies β.

Table 2: Parameters and Moments in Calibration

Parameter	Moment
γ: elasticity between consumption and housing-fertility	$\text{corr}(n_i, w_i)$
ρ: elasticity between housing and fertility	$\text{corr}(n_i, L_i)$
α: goods consumption share	average fertility n_i
β: housing rent share	aggregate rent-to-income ratio

Data Moments. We construct:

$$m^{data}(\gamma, \rho, \alpha, \beta) = \begin{pmatrix} \text{corr}(n_i, w_i) \\ \text{corr}(n_i, L_i) \\ \frac{1}{N}\sum_{i=1}^{N} n_i \\ \frac{\sum_{i=1}^{N} h_i L_i}{\sum_{i=1}^{N} w_i L_i} \end{pmatrix}. \tag{49}$$

Treatment of the Time Cost Parameter. By allowing χ_i to vary freely, we could match observed fertility exactly for any preference parameters, leaving them unidentified. To address this, we impose $\chi_i = 0.15$ for all regions during preference parameter calibration, following the average estimate in the literature. This forces the model to explain cross-sectional fertility variation through endogenous responses to wages and housing costs.

Calibration Algorithm.

Algorithm 1. *We use a Newton-Raphson algorithm with an outer loop updating preference parameters and an inner loop solving for equilibrium:*

- *Initial guess for* $(\gamma, \rho, \alpha, \beta)$. *Then the* ***inner loop****:*

  1. *Initial guess for* (A_i, B_i)
  2. *Compute* $L_i^e = (1 - \chi_i n_i) L_i$ *from the data*
  3. *Compute* (P_i, Ξ_i, C_i) *using Equations* (28), (26), *and* (24)
  4. *Update* A_i *by*
  $$A_i = \left[\frac{w_i^{\sigma} \left(L_i^e\right)^{1-\psi(\sigma-1)}}{\sum_{n=1}^{N} \tau_{in}^{1-\sigma} P_n^{\sigma} C_n L_n} \right]^{\frac{1}{\sigma-1}} \tag{50}$$
  5. *Compute* $V_i = \left[\alpha^{\frac{1}{\gamma}} C_i^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h_i^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n_i^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1} \frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}$
  6. *Update* B_i *by*
  $$B_i = \frac{g L_i}{\sum_{o=1}^{N} \frac{\left(\frac{1}{d_{oi}} V_i\right)^{\eta}}{\sum_{k=1}^{N} B_k^{(0)} \left(\frac{1}{d_{ok}} V_k\right)^{\eta}} n_o L_o} \tag{51}$$
  7. *Iterate until convergence*

- *Compute simulated fertility* n_i^{sim} *using Equation* (29)

- *Compute simulated per-capita housing* h_i^{sim} *using Equation* (27)

- *Compute simulated moments:*

$$m^{sim}(\gamma, \rho, \alpha, \beta) = \begin{pmatrix} corr(n_i^{sim}, w_i) \\ corr(n_i^{sim}, L_i) \\ \frac{1}{N} \sum_{i=1}^{N} n_i^{sim} \\ \frac{\sum_{i=1}^{N} h_i^{sim} L_i}{\sum_{i=1}^{N} w_i L_i} \end{pmatrix} \tag{52}$$

- *Let* $f(\gamma, \rho, \alpha, \beta) \equiv m^{sim} - m^{data}$ *and* J *be its Jacobian. Update:*

$$(\gamma^{new}, \rho^{new}, \alpha^{new}, \beta^{new})' = (\gamma, \rho, \alpha, \beta)' - J^{-1} \times f(\gamma, \rho, \alpha, \beta) \tag{53}$$

- *Iterate until convergence*

Calibration Results. Table 3 reports the calibrated values. The elasticity $\gamma = 0.91$ indicates complementarity between consumption and the housing-fertility composite. The elasticity $\rho = 0.49$ indicates strong complementarity between housing and children—this is central to our analysis. When housing becomes expensive, households cut back on both housing and children rather than substituting toward children. The consumption share is $\alpha = 0.66$, and the housing share within the composite is $\beta = 0.60$.

Table 3: Calibrated Preference Parameters

γ	ρ	α	β
0.9082	0.4866	0.6590	0.5960

4.4 Model Fit

Given calibrated parameters, we fit the model to exactly match observed wages, population, and fertility by recovering productivity A_i, amenities B_i, and time costs χ_i. Intuitively, productivity is identified by wages, amenities by population, and time costs by fertility.

Algorithm 2. *We use the following algorithm to recover* (A_i, B_i, χ_i) *that exactly match data on* (w_i, L_i, n_i)*:*

1. *Initial guess for* (A_i, B_i, χ_i)
2. *Solve steady-state equilibrium (Definition 3), obtaining* (w_i, L_i, n_i)
3. *Update:*

$$
\begin{aligned}
A_i &= A_i \left(\frac{w_i^{data}}{w_i} \right)^{\iota_A} \\
B_i &= B_i \left(\frac{L_i^{data} / \left(\sum_{k=1}^{N} L_k^{data} \right)}{L_i / \left(\sum_{k=1}^{N} L_k \right)} \right)^{\iota_B} \\
\chi_i &= \chi_i \left(\frac{n_i}{n_i^{data}} \right)^{\iota_\chi}
\end{aligned}
\tag{54}
$$

where ι_A*,* ι_B*,* ι_χ *govern adjustment speed*

4. Iterate until convergence

By construction, the calibrated model exactly matches observed wages, population, and fertility. To assess the model's explanatory power through economic fundamentals alone, Figure 2 plots model-predicted fertility under uniform χ_i against observed fertility. The slope of 0.66 (standard error 0.15) indicates that the model captures meaningful variation even without region-specific fertility shifters. The R^2 of 0.03 shows that economic fundamentals explain approximately 3 percent of fertility variance—demonstrating that the model's mechanisms have quantitative bite, while leaving room for non-economic factors captured in χ_i.

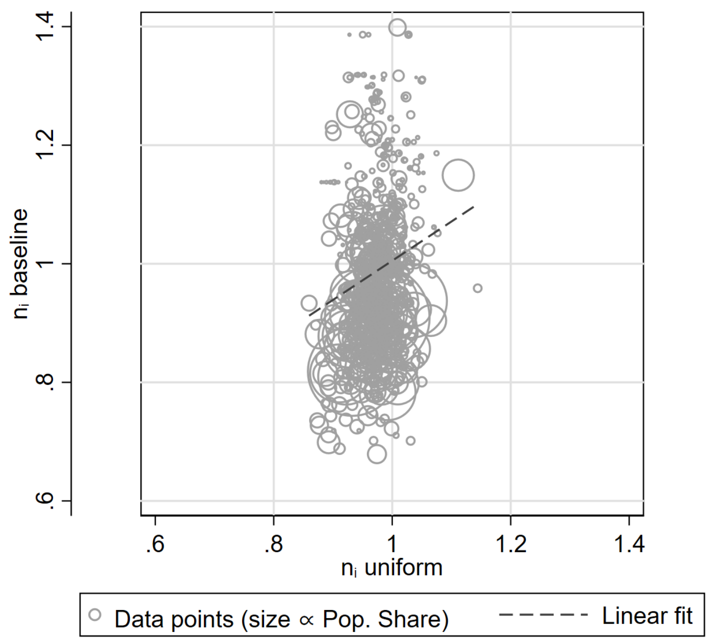


Figure 2: Model Fit: Fertility under Uniform Time Costs χ_i

Notes: "n_i baseline" refers to fertility rates in the baseline model (identical to data). "n_i uniform" refers to fertility rates under $\chi_i = \text{mean}(\chi_i)$ for all i. Slope: 0.66 (s.e. 0.15); $R^2 = 0.03$. Pop. Share refers to population share in the data.

We also regress the fertility rate, n_i, simulated under uniform χ_i, on simulated wages, population, and housing rents across space. Table 4 shows that, even absent cross-CZ variation in χ_i, the model reproduces the negative correlations between fertility and wages, population, and housing rents documented in Figure 1.

Figure 3 compares model-predicted housing rents to observed rents. The slope of 0.98 (standard error 0.03) indicates no systematic bias, and the R^2 of 0.61 shows that the model explains 61

Table 4: Correlations between Fertility, Wage, Population, and Housing Rent under Uniform Time Costs χ_i

	Dependent Variable: n_i under uniform χ_i		
	(1)	(2)	(3)
$\log(w_i)$ under uniform χ_i	-0.041^{***}		
	(0.008)		
$\log(L_i)$ under uniform χ_i		-0.004^{***}	
		(0.001)	
$\log(r_i)$ under uniform χ_i			-0.154^{***}
			(0.006)
Observations	741	741	741
R-squared	0.039	0.038	0.556

percent of rent variance. This close fit is important for counterfactual credibility, since housing costs are a key channel affecting fertility.

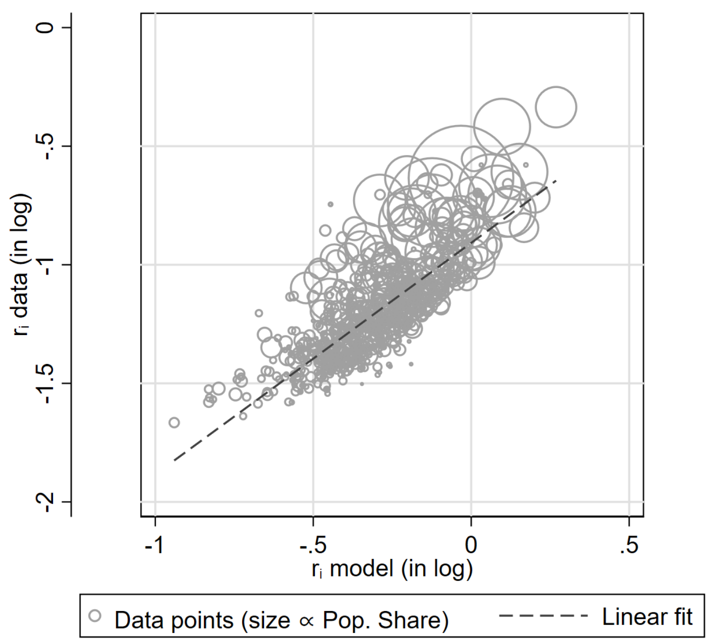
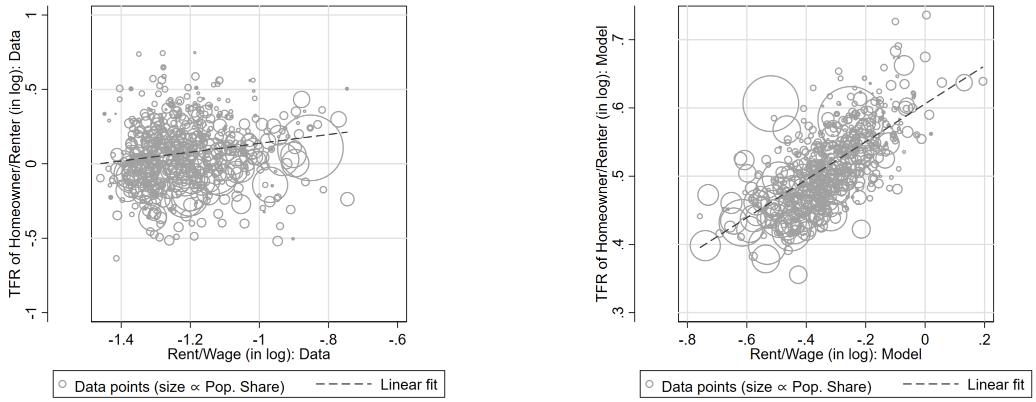


Figure 3: Model Fit: Housing Rents

Notes: "r_i data" refers to observed housing rents. "r_i model" refers to model-predicted rents. Slope: 0.98 (s.e. 0.03); $R^2 = 0.61$. Pop. Share refers to population share in the data.

In summary, the calibrated model matches observed wages, population, and fertility exactly,



(a) Data

(b) Model

Figure 4: Model Fit: Fertility of Homeowners and Renters

Notes: The fitted line coefficients (standard errors) in panels (a) and (b) are 0.30 (0.07) and 0.28 (0.01) respectively.

while delivering a close fit to housing rents. The estimated preference parameters indicate substantial complementarity between housing and fertility, providing a structural foundation for the negative correlation between housing costs and fertility.

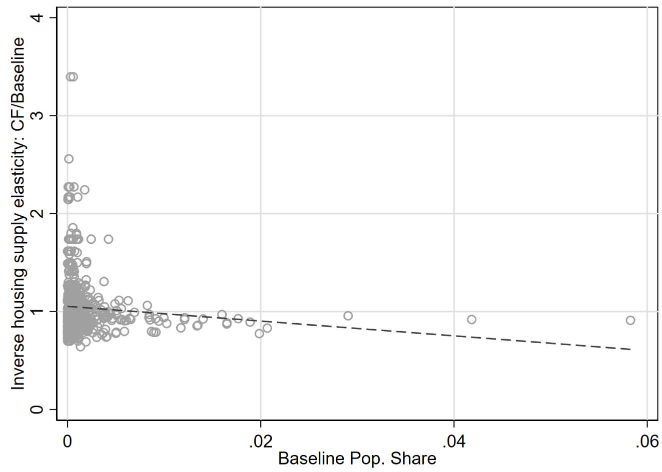
5 Counterfactual Analysis

5.1 Housing Supply Regulations over Space

We use the calibrated model to evaluate how housing supply regulations affects fertility, population distribution, and welfare over space. To this end, we equalize land-use regulations across all 741 commuting zones to the median. In particular, we shut down cross-sectional variation in land-use regulations by assigning every MSA the median value of the Wharton Residential Land Use Regulatory Index (WRLURI), while preserving each location's observed geographic fundamentals. Applying the structural coefficients from Saiz (2010), we recompute the inverse housing supply elasticity for each MSA under this uniform-regulation scenario and derive the counterfactual MSA-level elasticity as its reciprocal. We then map these counterfactual MSA elasticities onto 1990 Commuting Zones (CZs) using our baseline spatial matching procedure.

Figure 5 suggests that equalizing housing supply regulations to the median effectively relaxes

Figure 5: Changes in Housing Supply Elasticities (ξ_i): Equalizing Housing Supply Regulations to the Median



Notes: Baseline pop. share refers to $\frac{L_i}{\sum_{k=1}^{N} L_k}$ in our baseline model. "Inverse Housing Supply Elasticity: CF/Baseline" refers to $\xi_i^{CF}/\xi_i^{Baseline}$.

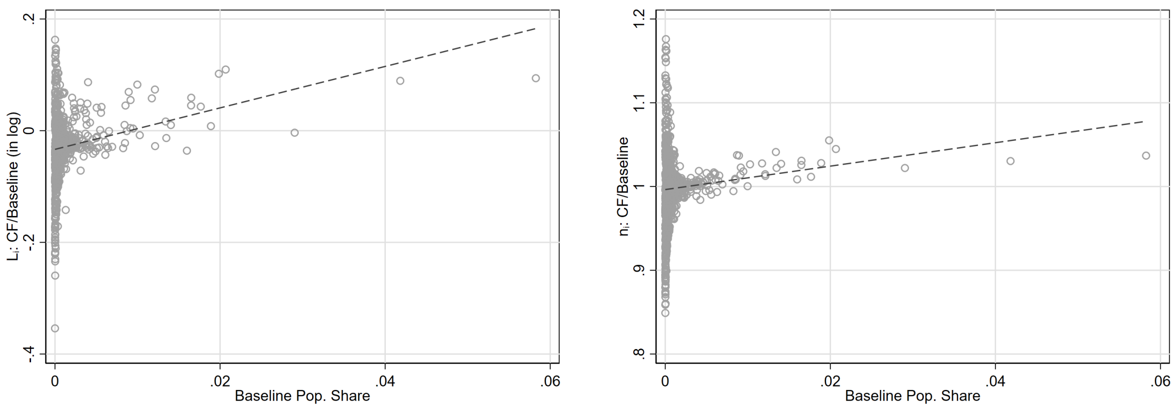
housing supply constraints in the most populous and productive locations, while tightening them in CZs with initially more permissive regulation.

Figure 6 displays the spatial distribution of changes in population and fertility following the equalization of regulations. Panel (a) shows that population increases disproportionately in CZs that were initially large. This reallocation reflects the core mechanism of our model: relaxing housing supply constraints in productive cities initially reduces rents, raises real wages, and thereby attracts migrants. The largest population gains accrue to CZs in the upper right of the figure—those with high initial population shares—while smaller CZs experience modest population declines as workers relocate toward now-more-affordable productive centers.

Panel (b) shows the corresponding pattern for fertility. Fertility increases in the large CZs where housing supply constraints have been relaxed. The mechanism operates through the housing-fertility complementarity documented in our calibration: with $\rho = 0.49$, housing and children are strong complements, so reducing housing costs raises demand for both. In the previously supply-constrained large CZs, the policy-induced rent decline translates directly into higher fertility through Equation (29). Meanwhile, fertility changes are more muted, and in some cases negative, in smaller CZs whose regulation is tightened by the policy.

The positive relationship between baseline population share and fertility gains visible in Panel (b) illustrates why this policy raises aggregate fertility rather than merely redistributing it. Unlike policies that attract households into low-fertility environments—thereby reducing aggregate fertility through compositional effects—equalizing regulation raises fertility precisely in the locations receiving population inflows.

Figure 6: Changes in Population Allocation and Fertility over Space: Equalizing Housing Supply Regulations to the Median



(a) Labor

(b) Fertility

Notes: Baseline pop. share refers to $\frac{L_i}{\sum_{k=1}^{N} L_k}$ in our baseline model. "L_i: CF/Baseline (in log)" refers to $\log\left(L_i^{CF}/L_i^{Baseline}\right)$. "$n_i$: CF/Baseline" refers to $n_i^{CF}/n_i^{Baseline}$.

Table 5 summarizes the aggregate and distributional effects of equalizing regulation:

The aggregate steady-state population growth g rises from 0.92 in the baseline to 0.93 under the counterfactual. By reducing housing costs in productive cities where fertility was previously depressed, the policy raises fertility in locations that account for a large share of the population, generating aggregate gains.

Table 5: Aggregate and Distributional Effects: Equalizing Regulation towards its Median

	g	$std\,(w_i/P_i)$	$std\,(r_i/P_i)$	$std\,(\log L_i)$
Baseline	0.9188	0.5366	0.3252	1.6328
CF	0.9288	0.5364	0.3464	1.6537

The distributional consequences reveal an interesting pattern. The standard deviation of real wages w_i/P_i across CZs falls from 0.5366 to 0.5364, indicating modest spatial convergence in living standards. This compression reflects the reallocation of workers toward high-wage locations:

as workers move to productive cities with now-relaxed housing constraints, the wage gap between high- and low-productivity locations narrows through labor market arbitrage.

In contrast, the standard deviation of real housing rents r_i/P_i rises from 0.33 to 0.35. This increased dispersion emerges because the policy affects CZs asymmetrically. In CZs where regulation was initially above the median, the policy lowers the inverse supply elasticity, moderating rent increases in response to population inflows. In CZs where regulation was initially below the median, the policy raises the inverse supply elasticity, making rents more responsive to demand changes. The net effect is greater spatial variation in real rents, even as real wage variation declines.

6 Conclusion

This paper develops a quantitative spatial equilibrium model with endogenous fertility to study the interaction between demographic dynamics and the spatial distribution of economic activity. We document that fertility rates are strongly negatively correlated with wages and housing costs across U.S. commuting zones, and we provide a structural interpretation of these patterns through a model in which households jointly choose location, consumption, housing, and fertility.

Our framework integrates household decisions in a multi-region economy with trade, migration, and heterogeneous housing supply. The key insight is that housing and children are complements in household preferences: when housing becomes expensive, households reduce both housing consumption and fertility. We estimate this complementarity to be substantial, with an elasticity of substitution between housing and fertility of 0.49. This parameter implies that housing supply constraints not only reduce welfare through higher living costs but also depress fertility, with consequences for long-run population dynamics.

The model yields several analytical results illuminating the mechanisms at work. Local fertility elasticities decompose into within-nest and cross-nest substitution effects plus income effects, with the relative magnitudes determined by the elasticities of substitution γ and ρ. Cross-place spillovers operate through trade, migration, and price channels, so that shocks in one region affect fertility throughout the economy. Endogenous fertility acts as a stabilizing force in spatial dynamics, accelerating convergence to steady state as regions receiving population inflows experience rent increases that moderate fertility and dampen further growth.

Our counterfactual analysis demonstrates the quantitative importance of housing supply constraints for demographic outcomes. Equalizing housing supply regulations at the median across commuting zones raises the aggregate steady-state population growth from 0.92 to 0.93. This effect arises because the policy raises fertility precisely in the large, productive commuting zones that account for substantial population shares. The intensive margin (higher fertility in supply-constrained locations) and extensive margin (population reallocation toward those locations) reinforce rather than offset each other. The policy also compresses the distribution of real wages across space while widening the distribution of real housing rents.

These findings carry implications for policy. First, housing supply reform may be a powerful lever for addressing demographic decline. By reducing housing costs in productive cities, such reforms raise fertility where it matters most for aggregate outcomes. Second, the efficiency-demography tradeoff emphasized in prior work is not inevitable. Policies that reduce the cost of living in productive cities can simultaneously improve allocative efficiency and raise aggregate fertility, provided they operate on costs directly rather than simply making cities more attractive along other dimensions. Third, our results complement the findings of Hsieh and Moretti (2019) on housing supply and spatial misallocation: the welfare costs of housing regulation include not only static efficiency losses but also dynamic demographic consequences that compound over time.

Several extensions would be valuable. Incorporating heterogeneity in worker skills would allow us to study how the efficiency-demography tradeoff varies across the skill distribution and how skill-biased spatial sorting affects aggregate fertility. Adding dynamics of human capital formation would connect our framework to the literature on growth and intergenerational mobility. Extending the model to include international migration would enable analysis of how immigration policy interacts with domestic demographic dynamics. Modeling education competition, as in Arkolakis et al. (2026), would allow us to study an additional channel through which urban density affects the cost of children. We leave these extensions for future work.

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A Theory

A.1 Proof of Proposition 1 and Corollary 1

We derive the fertility elasticities from the household's optimal fertility choice. The fertility demand function is:

$$n_i = \frac{1}{\chi_i w_i} \cdot \frac{(1-\beta)(\chi_i w_i)^{1-\rho}}{\Psi_i} \cdot \frac{(1-\alpha)\Psi_i^{\frac{1-\gamma}{1-\rho}}}{\Xi_i} \cdot (w_i + T_i), \tag{A.1}$$

where:

$$\Psi_i \equiv \beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}, \tag{A.2}$$

$$\Xi_i \equiv \alpha P_i^{1-\gamma} + (1-\alpha)\Psi_i^{\frac{1-\gamma}{1-\rho}}. \tag{A.3}$$

Step 1: Log-linearization of fertility demand.

Taking logs of (A.1):

$$\begin{aligned}\log n_i &= -\log(\chi_i w_i) + (1-\rho)\log(\chi_i w_i) - \log\Psi_i + \frac{1-\gamma}{1-\rho}\log\Psi_i - \log\Xi_i + \log(w_i+T_i) + \kappa \\ &= -\rho\log(\chi_i w_i) + \frac{\rho-\gamma}{1-\rho}\log\Psi_i - \log\Xi_i + \log(w_i+T_i) + \kappa, \end{aligned} \tag{A.4}$$

where $\kappa = \log[(1-\beta)(1-\alpha)]$ is a constant.

Step 2: Elasticities of Ψ_i and Ξ_i.

From the definition (A.2):

$$\frac{\partial \log\Psi_i}{\partial \log w_i} = \frac{(1-\beta)(1-\rho)(\chi_i w_i)^{1-\rho}}{\Psi_i} = (1-\rho)\tilde{\omega}_i^n, \tag{A.5}$$

$$\frac{\partial \log\Psi_i}{\partial \log r_i} = \frac{\beta(1-\rho)r_i^{1-\rho}}{\Psi_i} = (1-\rho)\tilde{\omega}_i^h, \tag{A.6}$$

$$\frac{\partial \log\Psi_i}{\partial \log \chi_i} = (1-\rho)\tilde{\omega}_i^n. \tag{A.7}$$

From the definition (A.3), using the chain rule:

$$\frac{\partial \log \Xi_i}{\partial \log \Psi_i} = \frac{(1-\alpha)\Psi_i^{\frac{1-\gamma}{1-\rho}}}{\Xi_i} \cdot \frac{1-\gamma}{1-\rho} = (1-\omega_i^C) \cdot \frac{1-\gamma}{1-\rho}, \tag{A.8}$$

where we used $1-\omega_i^C = (1-\alpha)\Psi_i^{\frac{1-\gamma}{1-\rho}}/\Xi_i$. Therefore:

$$\frac{\partial \log \Xi_i}{\partial \log w_i} = (1-\omega_i^C) \cdot \frac{1-\gamma}{1-\rho} \cdot (1-\rho)\tilde{\omega}_i^n = (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n, \tag{A.9}$$

$$\frac{\partial \log \Xi_i}{\partial \log r_i} = (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^h. \tag{A.10}$$

Step 3: Wage elasticity of fertility.

Differentiating (A.4) with respect to $\log w_i$, holding T_i fixed:

$$\begin{aligned}\varepsilon_{n,w} &= -\rho + \frac{\rho-\gamma}{1-\rho} \cdot (1-\rho)\tilde{\omega}_i^n - (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n + \frac{w_i}{w_i+T_i} \\ &= \underbrace{-\rho - (\gamma-\rho)\tilde{\omega}_i^n - (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n}_{\text{substitution effect}} + \underbrace{\frac{w_i}{w_i+T_i}}_{\text{income effect}}.\end{aligned} \tag{A.11}$$

The substitution effect can be rewritten to clarify its sign. Factoring out $\tilde{\omega}_i^n$ from the last two terms:

$$-\rho + \tilde{\omega}_i^n \left[-(\gamma-\rho) - (1-\gamma)(1-\omega_i^C)\right]. \tag{A.12}$$

If $\rho \le \gamma < 1$, then the substitution effect is negative.

Step 4: Rent elasticity of fertility.

Differentiating (A.4) with respect to $\log r_i$, holding T_i fixed:

$$\begin{aligned}\varepsilon_{n,r} &= \frac{\rho-\gamma}{1-\rho} \cdot (1-\rho)\tilde{\omega}_i^h - (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^h \\ &= \tilde{\omega}_i^h \left[-(\gamma-\rho) - (1-\gamma)(1-\omega_i^C)\right].\end{aligned} \tag{A.13}$$

It is straightforward that $\varepsilon_{n,r} < 0$ if $\rho \le \gamma < 1$.

Step 5: Time cost elasticity of fertility.

Differentiating (A.4) with respect to $\log \chi_i$:

$$\varepsilon_{n,\chi} = -\rho - (\gamma - \rho)\tilde{\omega}_i^n - (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n = \varepsilon_{n,w} - \frac{w_i}{w_i + T_i}. \tag{A.14}$$

This relationship holds because χ_i and w_i enter symmetrically in the opportunity cost $\chi_i w_i$, but only w_i affects income. $\square$

Step 6: Fertility Rates of Homeowners and Renters. Notice that the aggregate local fertility rate is a linear combination of fertility rates of homeowners and renters. Therefore, we have

$$\tilde{\zeta}_i d\log n_i^O + \left(1 - \tilde{\zeta}_i\right) d\log n_i^R = d\log n_i, \quad \tilde{\zeta}_i = \zeta_i \frac{n_i^O}{n_i^R}. \tag{A.15}$$

The ratio of n_i^O and n_i^R satisfies

$$\frac{n_i^O}{n_i^R} = \frac{w_i + \frac{1}{\zeta_i}T_i}{w_i} \quad \Rightarrow \quad d\log n_i^O - d\log n_i^R = \frac{\frac{1}{\zeta_i}T_i}{w_i + \frac{1}{\zeta_i}T_i}\varepsilon_{T,r}. \tag{A.16}$$

Combining two equations, we get the results in Corollary 1.

A.2 Proof of Proposition 2

Trade channel.

The price index in region i is:

$$P_i = \left[\sum_{k=1}^{N}\left(\tau_{ik} \cdot \frac{w_k}{A_k (L_k^e)^\psi}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}. \tag{A.17}$$

Define unit cost $c_k \equiv w_k / [A_k (L_k^e)^\psi]$. The expenditure share is:

$$\lambda_{ik} = \frac{(\tau_{ik} c_k)^{1-\sigma}}{P_i^{1-\sigma}}. \tag{A.18}$$

Taking logs: $(1-\sigma)\log P_i = \log\left[\sum_k (\tau_{ik} c_k)^{1-\sigma}\right]$.

Differentiating with respect to $\log A_j$, holding w_k and L^e_k fixed:

$$(1-\sigma)\frac{\partial \log P_i}{\partial \log A_j} = \frac{(1-\sigma)(\tau_{ij}c_j)^{1-\sigma}}{P_i^{1-\sigma}} \cdot \frac{\partial \log c_j}{\partial \log A_j} = (1-\sigma)\lambda_{ij}\cdot(-1). \tag{A.19}$$

Thus $\partial \log P_i / \partial \log A_j|_{w,L^e} = -\lambda_{ij}$.

In general equilibrium, log-linearizing:

$$\hat{P}_i = \sum_k \lambda_{ik}\hat{c}_k = \sum_k \lambda_{ik}(\hat{w}_k - \hat{A}_k - \psi \hat{L}^e_k). \tag{A.20}$$

Migration channel.

In steady state: $L_i = \sum_o m_{oi} n_o L_o$, where $m_{oi} = B_i(d_{oi}^{-1}V_i)^\eta / \Phi_o$.

Log-linearizing the migration probability:

$$\hat{m}_{oi} = \eta \hat{V}_i - \hat{\Phi}_o = \eta \hat{V}_i - \eta \sum_k m_{ok}\hat{V}_k = \eta(\hat{V}_i - \bar{\hat{V}}_o). \tag{A.21}$$

Log-linearizing population:

$$\hat{L}_i = \sum_o \frac{m_{oi}n_oL_o}{L_i}(\hat{m}_{oi} + \hat{n}_o + \hat{L}_o) = \sum_o \pi_{oi}\left[\eta(\hat{V}_i - \bar{\hat{V}}_o) + \hat{n}_o + \hat{L}_o\right]. \tag{A.22}$$

Housing price channel.

From $r_i = \bar{r}_i H_i^{\xi_i}$: $\log r_i = \log \bar{r}_i + \xi_i \log H_i$, so $\hat{r}_i = \xi_i \hat{H}_i$.

From market clearing $H_i = h_i L_i$: $\hat{H}_i = \hat{h}_i + \hat{L}_i$.

Combining: $\hat{r}_i = \xi_i(\hat{h}_i + \hat{L}_i)$. □

A.3 Proof of Proposition 3

The steady-state population in region j satisfies:

$$L_j = m_{jj}n_jL_j + \sum_{o \neq j} m_{oj}n_oL_o. \tag{A.23}$$

Define the natural retention rate $\mu_j \equiv m_{jj} n_j$ and net immigration $I_j \equiv \sum_{o \neq j} m_{oj} n_o L_o$. Then:

$$L_j = \frac{I_j}{1 - \mu_j}. \tag{A.24}$$

Log-linearizing:

$$\hat{L}_j = \hat{I}_j + \frac{\mu_j}{1 - \mu_j} \hat{\mu}_j = \hat{I}_j + \frac{\mu_j}{1 - \mu_j} (\hat{m}_{jj} + \hat{n}_j). \tag{A.25}$$

With exogenous fertility ($\hat{n}_j = 0$):

$$\hat{L}_j^{SS}\big|_{\text{exo } n} = \hat{I}_j + \frac{\mu_j}{1 - \mu_j} \hat{m}_{jj}. \tag{A.26}$$

With endogenous fertility:

$$\hat{L}_j^{SS}\big|_{\text{endo } n} = \hat{I}_j + \frac{\mu_j}{1 - \mu_j} (\hat{m}_{jj} + \hat{n}_j). \tag{A.27}$$

The difference is:

$$\hat{L}_j^{SS}\big|_{\text{endo } n} - \hat{L}_j^{SS}\big|_{\text{exo } n} = \frac{\mu_j}{1 - \mu_j} \hat{n}_j < 0, \tag{A.28}$$

since $\hat{n}_j < 0$ (productivity shocks raise wages and rents, reducing fertility) and $\mu_j/(1 - \mu_j) > 0$. $\square$

A.4 Proof of Proposition 4

Aggregate fertility is $\bar{n} = \sum_i n_i s_i$ where $s_i = L_i/\bar{L}$ and $\sum_i s_i = 1$.

Taking the total differential:

$$d\bar{n} = \sum_i s_i \, dn_i + \sum_i n_i \, ds_i. \tag{A.29}$$

Since $\sum_i ds_i = 0$:

$$\sum_i n_i \, ds_i = \sum_i n_i \, ds_i - \bar{n} \sum_i ds_i = \sum_i (n_i - \bar{n}) \, ds_i. \tag{A.30}$$

For log changes, $ds_i = s_i(\hat{L}_i - \hat{\bar{L}})$ where $\hat{\bar{L}} = \sum_k s_k \hat{L}_k$. The extensive margin:

$$\sum_i (n_i - \bar{n}) s_i (\hat{L}_i - \hat{\bar{L}}) = \sum_i (n_i - \bar{n}) s_i \hat{L}_i = \mathrm{Cov}_s(n_i, \hat{L}_i), \tag{A.31}$$

using $\sum_i (n_i - \bar{n}) s_i = 0$.

Dividing by $\bar{n}$:

$$\hat{\bar{n}} = \sum_i \frac{s_i n_i}{\bar{n}} \hat{n}_i + \frac{1}{\bar{n}} \mathrm{Cov}_s(n_i, \hat{L}_i). \tag{A.32}$$

□

B Quantification

B.1 Data Construction

This appendix describes the construction of the dataset used in the calibration. The unit of observation is a commuting zone (CZ), defined following Autor and Dorn (2013).

B.1.1 Demographic and Economic Variables

We draw individual-level data from the IPUMS USA 5 percent sample of the 2000 Census (Ruggles et al., 2025). IPUMS reports geographic identifiers at the Public Use Microdata Area (PUMA) level. We aggregate all variables to the CZ level using the PUMA-to-CZ crosswalks from Autor and Dorn (2013).

Wages. We compute the average annual wage for individuals aged 16–60 within each CZ.

Fertility. We construct a CZ-level fertility rate for women aged 15–45. We use the Census question on the age of the youngest own child in the household. A woman is classified as having had a birth in the past year if the age of her youngest child is zero. The fertility rate is the ratio of such women to all women aged 15–45 in the CZ.

Housing Rent. We compute the average monthly gross rent at the CZ level, restricting the sample to housing units with cash rent.

Per-Capita Housing. We measure per-capita housing as the total number of bedrooms divided by CZ population.

Population. CZ population is the sum of all individuals enumerated in the CZ.

B.1.2 Migration Flows

We obtain county-to-county migration flows from the Census 2000 Special Tabulation on Geographic Mobility.[1] These tabulations record the number of individuals residing in a given county in 2000 who resided in a different county in 1995. We aggregate these flows to the CZ level using the county-to-CZ crosswalk to obtain a CZ-to-CZ migration matrix L_{oi}.

B.1.3 Trade Flows

Bilateral trade flows across CZs are not directly observed. We construct them in three steps. First, we obtain state-to-state commodity flows from the Commodity Flow Survey and compute great-circle distances between state centroids. Second, we estimate a gravity equation at the state level to recover the distance elasticity of trade, following Ramondo et al. (2016). Third, we compute great-circle distances between CZ centroids and apply the estimated gravity coefficients to construct the CZ-to-CZ trade share matrix λ_{in}.

B.1.4 Housing Supply Elasticity

We combine CZ-level housing data from IPUMS 2000 with the MSA-level housing supply elasticity estimates of Saiz (2010). The matching proceeds in three steps. First, we match IPUMS MSA codes to the MSA identifiers in Saiz (2010), supplemented by manual matching for codes that do not link automatically. Second, for the remaining 30–40 unmatched MSAs, we impute the elasticity using the population-weighted average elasticity of the corresponding state. Third, for

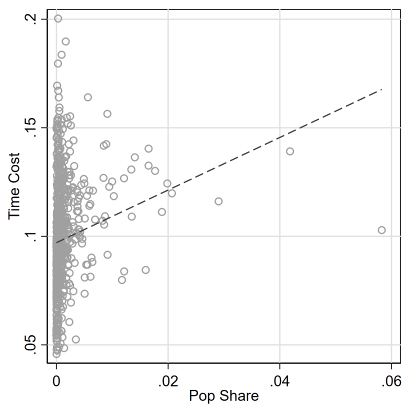
[1] `https://www.census.gov/data/tables/2000/demo/geographic-mobility/county-to-county-migration-flows.html`.

counties not belonging to any MSA, we assign the national population-weighted average elasticity. We then aggregate county-level elasticities to the CZ level using population weights.

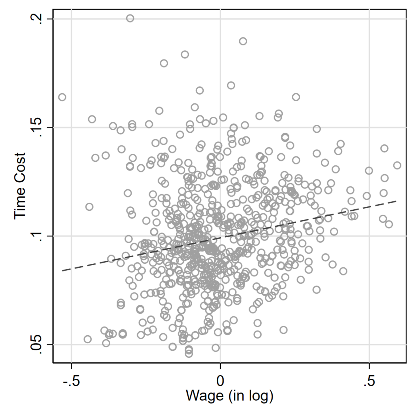
B.2 Calibration

B.2.1 Model Fit

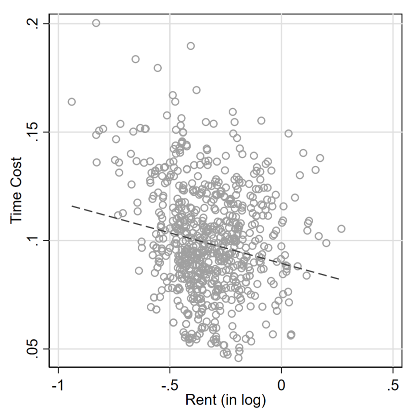
Figure A.1: Time Cost (χ_i) that Exactly Fit Fertility Rate (n_i) in the Data



(a) Population Share



(b) Wage



(c) Housing Rent

Notes: The fitted line coefficients (standard errors) in panels (a)–(c) are 1.21 (0.39), 0.03 (0.005), and −0.03 (0.006), respectively.